*KINGDOM OF SAUDI ARABIA*
*IMAM MOHAMMAD IBN SAUD ISLAMIC UNIVERSITY*
*COLLEGE OF COMPUTER AND INFORMATION SCIENCES*
*INFORMATION SYSTEMS DEPARTMENT*
*BACHELOR IN INFORMATION SYSTEMS*

المملكة العربية السعودية
جامعة الإمام محمد بن سعود الإسلامية
كلية علوم الحاسب والمعلومات
قسم نظم المعلومات
بكالوريوس نظم المعلومات

# *SYLLABUS*

## IS1356: Data Mining

| CREDIT HOURS | 3 credit hours (Lectures: 3) |
|---|---|

| PREREQUISITE | IS1340 |
|---|---|

| **Instructor:** |
|---|
| **Contact information and office hours** |
| **Office No:** To be announced (TBA)<br>**Office Hours: TBA**<br>**E-mail: _____@imamu.edu.sa** |

| COURSE DESCRIPTION |
|---|
| The aim of the course is to introduce students the knowledge of the importance, and benefits of data mining and its application. They will also gain a good understanding of the different data mining algorithms and the commonly used tools and techniques. |

| | COURSE LEARNING OUTCOMES (CLOs) | Aligned SOs |
|---|---|---|
| 1 | **Knowledge and Understanding** | |
| 1.1 | Recognize data mining importance, problems, and their solutions. | 1(I) |
| 1.2 | Outline data mining algorithms and its application | 1(I) |
| 1.3 | | |
| 1.4 | | |
| 1.5 | | |
| **2** | **Skills :** | |
| 2.1 | Create models using commonly used tools and techniques of data mining. | 2(I) |
| 2.2 | Prepare data after pre-processing before application of data mining algorithms. | 2(I) |

| | | |
|---|---|---|
| 2.3 | Analyze the results obtained from the data mining algorithms. | 2(I) |
| 2.4 | | |
| 2.5 | | |
| **3** | **Values:** | |
| 3.1 | Function effectively on teams to accomplish a common goal. | 5(I) |
| 3.2 | Present a topic in a compelling manner. | 3(I) |
| 3.3 | | |
| 3.4 | | |
| 3.5 | | |

| **TEACHING Strategies** |
|---|
| Lectures<br>Self-Learning |

| No | List of Topics | Contact Hours | Self-Learning |
|---|---|---|---|
| **1** | Topic 01- Introduction to Data Mining | *3* | |
| **2** | Topic 02 - Data Pre-processing | *3* | |
| **3** | Topic 03 - Classification - Nearest Neighbor | *3* | |
| **4** | Topic 04 - Classification - Naive Bayesian | *3* | |
| **5** | Topic 04 - Classification - Naive Bayesian Examples | *3* | *2* |
| **6** | Topic 05 - Classification - Decision Trees | *3* | |
| **7** | Topic 05 - Classification - Decision Tree Examples | *3* | *1* |
| **8** | Topic 06 - Classifier Performance | *3* | |
| **9** | Topic 06 - Classifier Performance Examples | *3* | *1* |
| **10** | Topic 07 –Data Mining Tools: Weka | *3* | |
| **11** | Topic 08 - ARM - Apriori Algorithm | *3* | |
| **12** | Topic 08 - ARM - Apriori Algorithm Examples | *3* | *2* |
| **Total** | | **36** | **6** |

| **TEXT BOOK** |
|---|
| M. Bramer, Principles of Data Mining,3rd edition, Springer Verlag, 2016<br><br>ISBN: 978-1-4471-7307-6 |

| **REFERENCES** |
|---|
| |

H. Witten, F. Eibe, and M. A. Hall, Data Mining: Practical Machine Learning Tools and Techniques, 3rd ed. Morgan Kaufman, 2011, ISBN: 978-0123748560
J. Han, M. Kamber and J. Pei, Data Mining: Concepts and Techniques, 3rd ed. Morgan Kaufman, Elsevier, 2012, ISBN: 978-0123814791
Pang-Ning Tan, Michael Steinbach, Vipin Kumar. Introduction to Data Mining, 1st ed. Pearson, 2005, ISBN: 978-0321321367

| | Course Assessment Methods | | |
|---|---|---|---|
| **No** | **Assessment Method** | **Due Week** | **% of Total Assessment** |
| 1 | **Quiz** | 4 | 10 |
| 2 | **Assignment** | 9 | 10 |
| 3 | **Midterm** | 7 | 20 |
| 4 | **Project / Lab Exam** | 11 | 20 |
| 5 | **Final Exam** | 13 | 40 |